HW2: Classification

Name: 莊明儒 Student ID: 111024513

due on 10/25 (Tue) 9am

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Your analysis should include the following:

- Exploratory data analysis: simple summaries with plots or tables
- Performing the classification task based on
  - logistic regression
  - discriminate analysis
  - nearest neighbor
- Report the performance of your classifiers and compare them among different methods.
- Make your conclusions on data contents.

1 Data Source

```
library(mlbench) #install package first!!
library(corrplot)
```

2 Problem 1: Wisconsin Breast Cancer Data

These data consist of 699 observations on 10 variables, one being "ID" variable, 9 being ordered or nominal variables, and 1 target class (benign or malignant). The objective is to classify each example into benign class or malignant class. More data descriptions can be found by typing `help(BreastCancer)` in r command.

Note that:

- There are 16 NA's in variable **Bare.nuclei**. If you don't know how to impute data, you may only use the observations with complete data, or you may drop this variable from your analysis.
- There are high correlations between all input variables, which should be aware of during modeling and data interpretation.

```
data(BreastCancer)
head(BreastCancer)
```

```
##        Id Cl.thickness Cell.size Cell.shape Marg.adhesion Epith.c.size
## 1 1000025            5         1          1             1            2
## 2 1002945            5         4          4             5            7
## 3 1015425            3         1          1             1            2
## 4 1016277            6         8          8             1            3
## 5 1017023            4         1          1             3            2
## 6 1017122            8        10         10             8            7
##   Bare.nuclei Bl.cromatin Normal.nucleoli Mitoses     Class
## 1           1           3               1       1    benign
## 2          10           3               2       1    benign
## 3           2           3               1       1    benign
## 4           4           3               7       1    benign
## 5           1           3               1       1    benign
## 6          10           9               7       1 malignant
```

```
dim(BreastCancer)
```

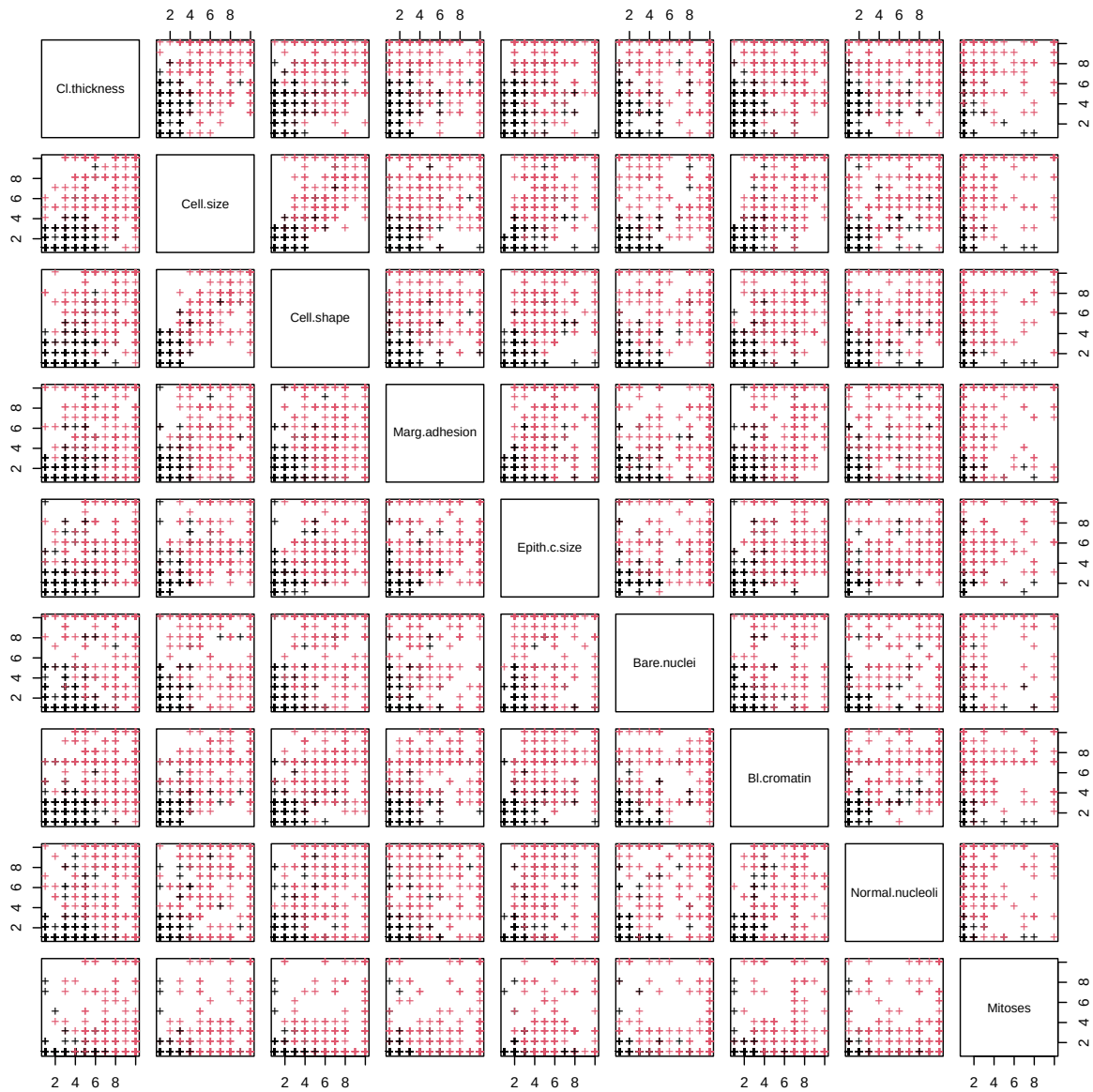
```
## [1] 699  11
```

```
#help(BreastCancer)

#make variables numeric (remove variable:ID) and save the data as a dataframe object
dat1 = matrix(as.numeric(as.matrix(BreastCancer[,2:10])), 699, 9)
dat1 = data.frame(dat1)
colnames(dat1) <- colnames(BreastCancer)[2:10]
head(dat1)
```

```
##   Cl.thickness Cell.size Cell.shape Marg.adhesion Epith.c.size Bare.nuclei
## 1            5         1          1             1            2           1
## 2            5         4          4             5            7          10
## 3            3         1          1             1            2           2
## 4            6         8          8             1            3           4
## 5            4         1          1             3            2           1
## 6            8        10         10             8            7          10
##   Bl.cromatin Normal.nucleoli Mitoses
## 1           3               1       1
```

```
## 2            3                2        1
## 3            3                1        1
## 4            3                7        1
## 5            3                1        1
## 6            9                7        1
```

```
dat1$case = as.numeric(BreastCancer$Class=="malignant")

pairs(dat1[,1:9], col=as.factor(dat1[,10]), pch="+")
```




```
round(cor(dat1, use="pairwise.complete.obs"),2) #handling data with NA
```

```
##                 Cl.thickness Cell.size Cell.shape Marg.adhesion Epith.c.size
```

```
## Cl.thickness            1.00      0.64       0.65          0.49         0.52
## Cell.size               0.64      1.00       0.91          0.71         0.75
## Cell.shape              0.65      0.91       1.00          0.68         0.72
## Marg.adhesion           0.49      0.71       0.68          1.00         0.60
## Epith.c.size            0.52      0.75       0.72          0.60         1.00
## Bare.nuclei             0.59      0.69       0.71          0.67         0.59
## Bl.cromatin             0.56      0.76       0.74          0.67         0.62
## Normal.nucleoli         0.54      0.72       0.72          0.60         0.63
## Mitoses                 0.35      0.46       0.44          0.42         0.48
## case                    0.72      0.82       0.82          0.70         0.68
##                 Bare.nuclei Bl.cromatin Normal.nucleoli Mitoses case
## Cl.thickness           0.59        0.56            0.54    0.35 0.72
## Cell.size              0.69        0.76            0.72    0.46 0.82
## Cell.shape             0.71        0.74            0.72    0.44 0.82
## Marg.adhesion          0.67        0.67            0.60    0.42 0.70
## Epith.c.size           0.59        0.62            0.63    0.48 0.68
## Bare.nuclei            1.00        0.68            0.58    0.34 0.82
## Bl.cromatin            0.68        1.00            0.67    0.34 0.76
## Normal.nucleoli        0.58        0.67            1.00    0.43 0.71
## Mitoses                0.34        0.34            0.43    1.00 0.42
## case                   0.82        0.76            0.71    0.42 1.00
```

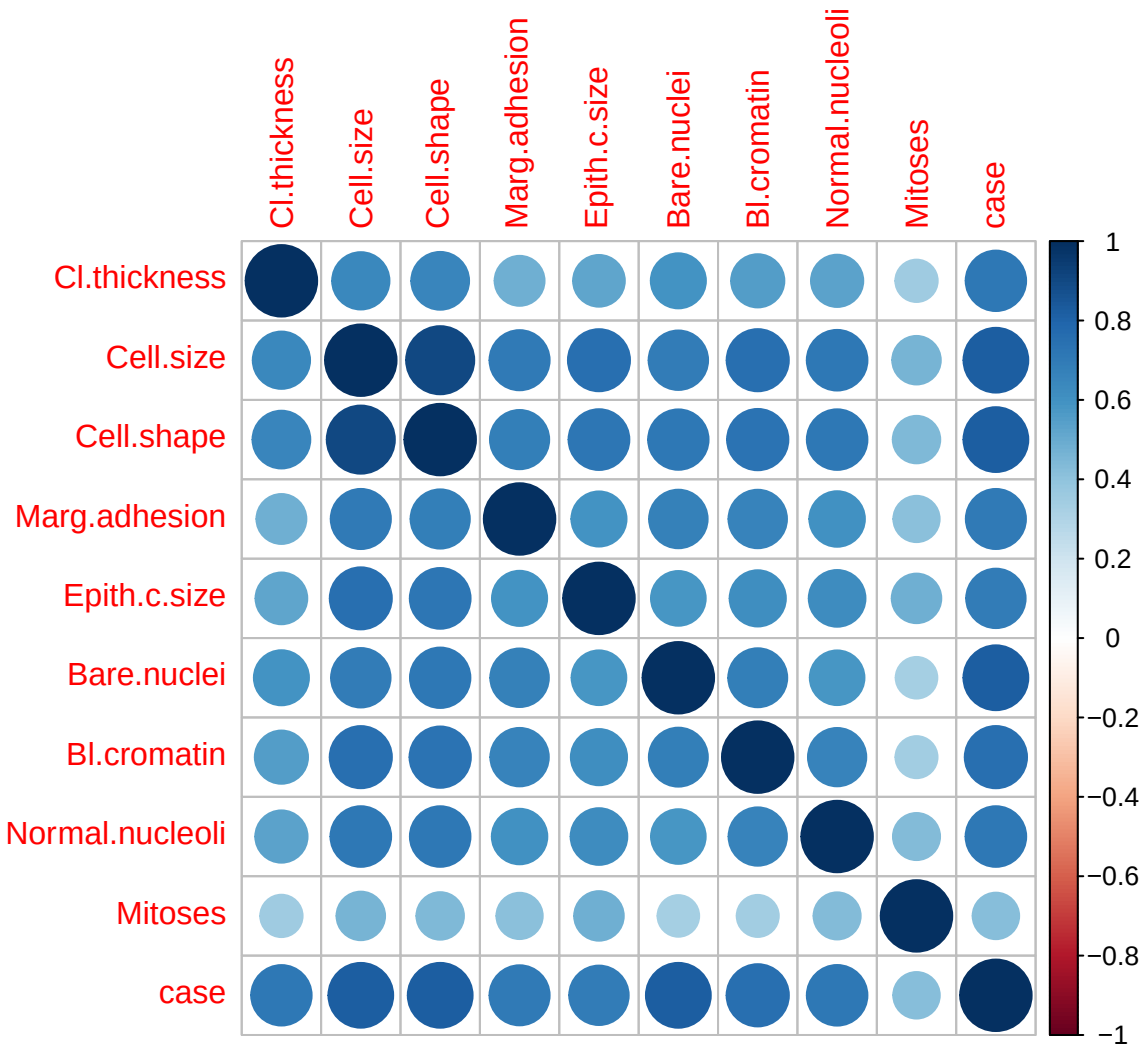
```r
#remove missing data (NA)
dat1 = na.omit(dat1)
dim(dat1) #check data dimension
```

```
## [1] 683  10
```

```r
#view variable correlations:
round(cor(dat1),2)
```

```
##                 Cl.thickness Cell.size Cell.shape Marg.adhesion Epith.c.size
## Cl.thickness            1.00      0.64       0.65          0.49         0.52
## Cell.size               0.64      1.00       0.91          0.71         0.75
## Cell.shape              0.65      0.91       1.00          0.69         0.72
## Marg.adhesion           0.49      0.71       0.69          1.00         0.59
## Epith.c.size            0.52      0.75       0.72          0.59         1.00
## Bare.nuclei             0.59      0.69       0.71          0.67         0.59
## Bl.cromatin             0.55      0.76       0.74          0.67         0.62
## Normal.nucleoli         0.53      0.72       0.72          0.60         0.63
## Mitoses                 0.35      0.46       0.44          0.42         0.48
## case                    0.71      0.82       0.82          0.71         0.69
##                 Bare.nuclei Bl.cromatin Normal.nucleoli Mitoses case
## Cl.thickness           0.59        0.55            0.53    0.35 0.71
## Cell.size              0.69        0.76            0.72    0.46 0.82
## Cell.shape             0.71        0.74            0.72    0.44 0.82
## Marg.adhesion          0.67        0.67            0.60    0.42 0.71
## Epith.c.size           0.59        0.62            0.63    0.48 0.69
## Bare.nuclei            1.00        0.68            0.58    0.34 0.82
## Bl.cromatin            0.68        1.00            0.67    0.35 0.76
## Normal.nucleoli        0.58        0.67            1.00    0.43 0.72
## Mitoses                0.34        0.35            0.43    1.00 0.42
## case                   0.82        0.76            0.72    0.42 1.00
```

```
corrplot(cor(dat1))
```



2.1 Exploratory Data Analysis

2.1.1 Data Description

This is a dataset consisting of 683 observations on 11 variables. One of the variables is ID, which is removed during the analysis. As for the rest of the variables, they are all quantitatively discrete. 16 missing values are present in the data. For simplicity, they are omitted directly.

A. As discrete variables

Table 1, 2, 3, 4 and 5 summarize the number of observations for each of the discrete variables (after omission of the missing values).

```
count = function(var1, var2, varname1, varname2){
  counts = t(data.frame(c(table(var1), table(var2))))
  rownames(counts) = "counts"
  header = c(1,length(names(table(var1))),length(names(table(var2))))
  header = header[header != 0]
  names(header) = c(" ", varname1, varname2)
  kable(counts, format = "latex", longtable = TRUE, booktabs = TRUE,
    col.names = c(names(table(var1)), names(table(var2))),
    align = "c", caption = paste("Summary for", varname1, "and", varname2), escape = FALSE
    ) %>%
    kable_styling(latex_options = c("striped")) %>%
    add_header_above(header)
}
cols = colnames(dat1)
count(dat1$Cl.thickness, dat1$Cell.size, cols[1], cols[2])
```

Table 1: Summary for Cl.thickness and Cell.size

	Cl.thickness										Cell.size									
	1	2	3	4	5	6	7	8	9	10	1	2	3	4	5	6	7	8	9	10
counts	139	50	104	79	128	33	23	44	14	69	373	45	52	38	30	25	19	28	6	67

```
count(dat1$Cell.shape, dat1$Marg.adhesion, cols[3], cols[4])
```

Table 2: Summary for Cell.shape and Marg.adhesion

	Cell.shape										Marg.adhesion									
	1	2	3	4	5	6	7	8	9	10	1	2	3	4	5	6	7	8	9	10
counts	346	58	53	43	32	29	30	27	7	58	393	58	58	33	23	21	13	25	4	55

```
count(dat1$Epith.c.size, dat1$Bare.nuclei, cols[5], cols[6])
```

Table 3: Summary for Epith.c.size and Bare.nuclei

	Epith.c.size	Bare.nuclei

	1	2	3	4	5	6	7	8	9	10	1	2	3	4	5	6	7	8	9	10
counts	44	376	71	48	39	40	11	21	2	31	402	30	28	19	30	4	8	21	9	132

```
count(dat1$Bl.cromatin, dat1$Normal.nucleoli, cols[7], cols[8])
```

Table 4: Summary for Bl.cromatin and Normal.nucleoli

	Bl.cromatin										Normal.nucleoli									
	1	2	3	4	5	6	7	8	9	10	1	2	3	4	5	6	7	8	9	10
counts	150	160	161	39	34	9	71	28	11	20	432	36	42	18	19	22	16	23	15	60

```
count(dat1$Mitoses, dat1$case, cols[9], cols[10])
```

Table 5: Summary for Mitoses and case

	Mitoses									case	
	1	2	3	4	5	6	7	8	10	0	1
counts	563	35	33	12	6	3	9	8	14	444	239

According to the tables above, several observations are summarized for each of the variables as follows.

1. CL.thickness (Clump thickness)

There is big variation of counts among the levels. As one can see, most of the subjects have their clump thickness at level 1, 3 and 5. Overall, CL.thickness seems to have a right-skewed distribution.

2. Cell.size (Uniformity of Cell Size)

Most of the subjects have their uniformity of cell size at level 1. Overall, Cell.size seems to have a strongly right-skewed distribution.

3. Cell.shape (Uniformity of Cell Shape)

Most of the subjects have their uniformity of cell shape at level 1. Overall, Cell.shape seems to have a strongly right-skewed distribution.

4. Marg.adhesion (Marginal Adhesion)

Most of the subjects have their marginal adhesion at level 1. Overall, Marg.adhesion seems to have a strongly right-skewed distribution.

5. Epith.c.size (Single Epithelial Cell Size)

Most of the subjects have their single epithelial cell size at level 2. Overall, Epith.c.size seems to have a strongly right-skewed distribution.

6. Bare.nuclei (Bare Nuclei)

Most of the subjects have their number of bare nuclei at level 1. Overall, Bare.nuclei seems to have a strongly right-skewed distribution.

7. Bl.cromatin (Bland Chromatin)

Most of the subjects have their number of bland chromatin at level 1, 2 and 3. Overall, Bl.cromatin seems to have a right-skewed distribution.

8. Normal.nucleoli (Normal Nucleoli)

Most of the subjects have their number of normal nucleoli at level 1. Overall, Normal.nucleoli seems to have a strongly right-skewed distribution.

9. Mitoses

Most of the subjects have their number of times of mitoses at level 1. Overall, Mitoses seems to have a strongly right-skewed distribution.

10 Case (Indicator variable for the cancer)

The number of subjects who are classified as benign class is about twice of that of those who are classified as malignant class. This fact may implies that the data is collected from a case control study

B. As quantitative variables

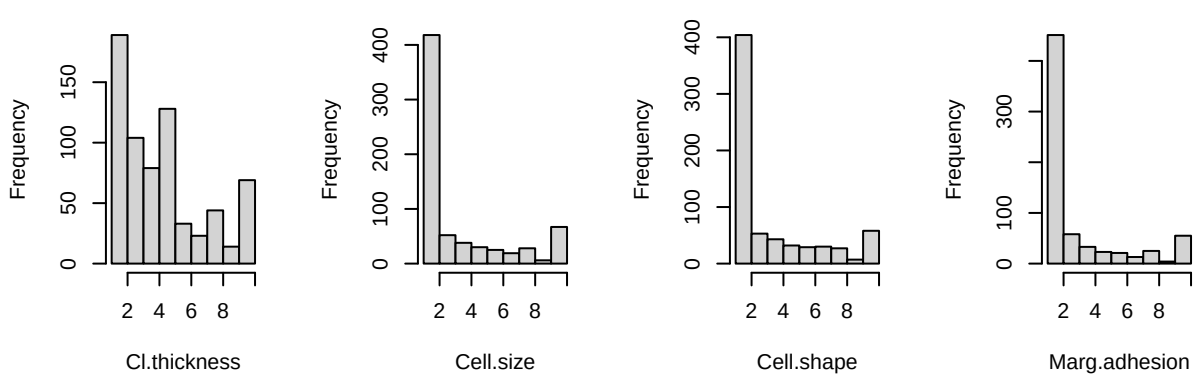
Due to the quantitative nature of the variables, it may also be useful to look at the five-number-summary and the mean of each variable. In order to understand the details of the distributions, histograms are also plotted.

```
summ = sapply(dat1[,1:9], summary)
kable(round(summ[,1:4], 2), format = "latex", longtable = TRUE, booktabs = TRUE,
  align = "c", caption = "Five-number-summary for each variable", escape = FALSE, linesep = ""
  ) %>%
  kable_styling(latex_options = c("striped"))
```

Table 6: Five-number-summary for each variable

	Cl.thickness	Cell.size	Cell.shape	Marg.adhesion
Min.	1.00	1.00	1.00	1.00
1st Qu.	2.00	1.00	1.00	1.00
Median	4.00	1.00	1.00	1.00
Mean	4.44	3.15	3.22	2.83
3rd Qu.	6.00	5.00	5.00	4.00
Max.	10.00	10.00	10.00	10.00

```
par(mfrow = c(1,4))
for(i in 1:4){
  hist(dat1[,i], main = "", xlab = cols[i])
}
```

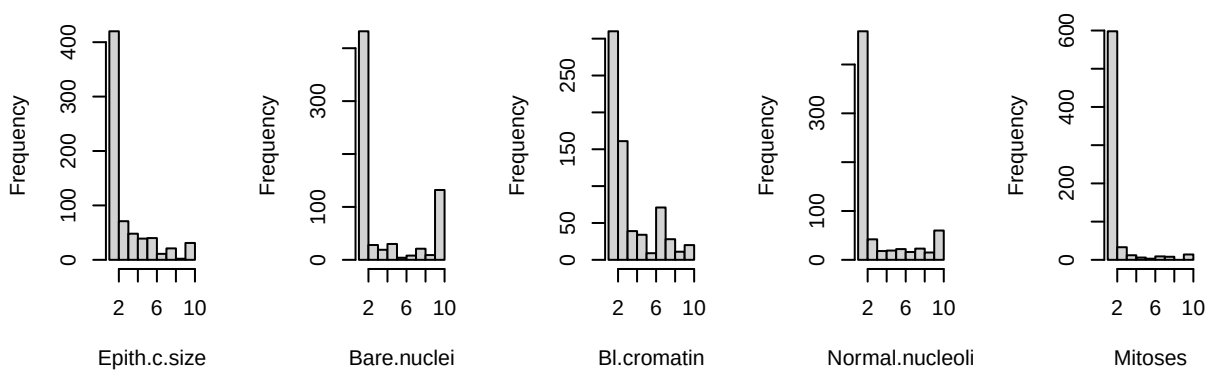


```
kable(round(summ[,5:9], 2), format = "latex", longtable = TRUE, booktabs = TRUE,
  align = "c", caption = "Five-number-summary for each variable", escape = FALSE, linesep = ""
  )  %>%
  kable_styling(latex_options = c("striped"))
```

Table 7: Five-number-summary for each variable

	Epith.c.size	Bare.nuclei	Bl.cromatin	Normal.nucleoli	Mitoses
Min.	1.00	1.00	1.00	1.00	1.0
1st Qu.	2.00	1.00	2.00	1.00	1.0
Median	2.00	1.00	3.00	1.00	1.0
Mean	3.23	3.54	3.45	2.87	1.6
3rd Qu.	4.00	6.00	5.00	4.00	1.0
Max.	10.00	10.00	10.00	10.00	10.0

```
par(mfrow = c(1,5))
for(i in 5:9){
  hist(dat1[,i], main = "", xlab = cols[i])
}
```



The difference between the median and the mean shown in Table 6 and 7 suggest right-skewed distribution for each variable. This is a consistent observation as reported previously. Moreover, as shown in the histograms, it seems that all the variables except for Mitoses have bimodal distributions, which may be resulted from the distinct distributions under the two classes.

2.1.2 Data Feature

To explore the basic relationships among the variables, variable visualization and correlation analysis are first conducted as follows.

```
dat1$case = factor(dat1$case, levels = 0:1)
ggpairs(dat1, columns = 1:9, ggplot2::aes(colour=case, alpha = 0.5),
        upper = list(continuous = wrap("cor", size = 2)),
        lower = list(continuous = wrap("points", size=0.2))) %>%
        + theme(strip.placement = "outside", text = element_text(size = 8))
```

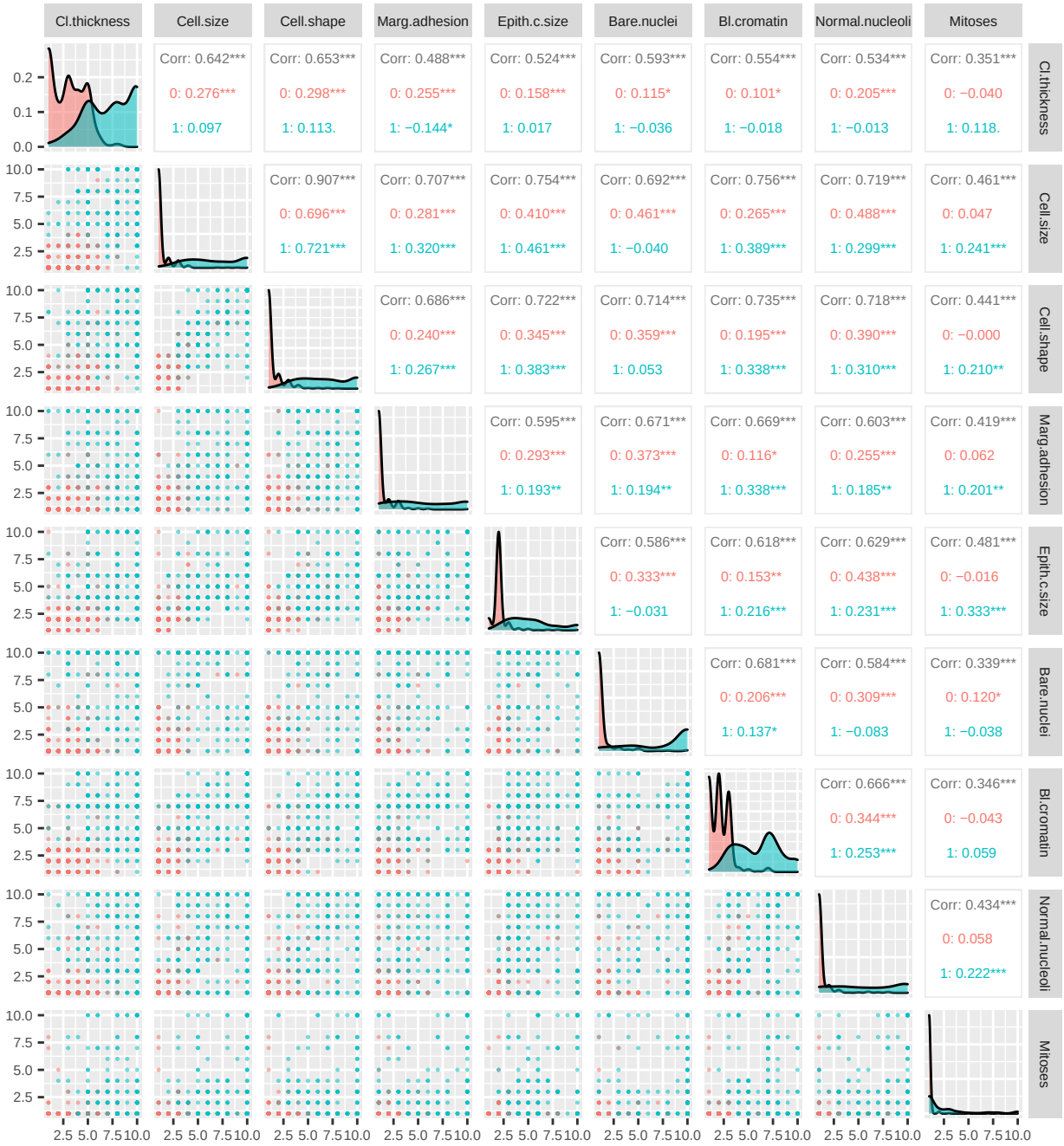


Figure 1: Summary for the relationships among the variables

In Figure 1 , the classes (benign and malignant) are discriminated by different colors, where red denotes for the benign and blue denotes for the malignant. From the diagonal (the histograms), it seems that all the variables may be useful in predicting the class in the sense that each of the variables has quite different distributions under the two classes. Several interesting findings are further listed as follows, which may be useful when fitting a model for the prediction problem in this case.

1. When fitting logistic regression

The lower triangle part (the scatter plots) of the figure shows that the effect of a variable on the probability

of being a certain class seems to depend on the other variable. For example, it seems that the increase on Cl.thickness seems to decrease the probability of benign class given that Cell.size is small. However, such effect does not seem to be obvious as Cell.size becomes large. Such fact suggests that there may be some additional benefit in terms of prediction performance if some of the interaction terms are included when fitting logistic regression for this problem.

2. When doing discriminant analysis

The histograms and scatter plots of the figure though suggest that multivariate normalily assumption may not be very suitable for the variables under the benign and malignant class, it should be still worth trying linear discriminant analysis which requires normality assumption. The scatter plots also suggest that the covariance of the variables does not seem to be identical under the two classes, implying that quadratic discriminant analysis may also be a good option for this problem. Moreover, since the number of variables seems to be large here, naive Bayes classifier may also provide good performance of prediction.

3. When doing K Nearest neighbor

The scatter plots of the figure shows that one may not need a very broken and tortuous boundary to separate the classes well, implying that the number of nearest neighbors k should not be selected as too small. Moreover, since the feature space is high dimensional, it seems to be quite difficult to choose an optimal k by inspection. Considering that the sample size seems to be large enough, cross validation may be a good way to select an appropriate number of k.

2.2 Performing the classification task

A. Train-test splitting

To evaluate the test mean squared error, the data is splitted into a training set and a testing set. About 90% of the data (615 observations) are randomly determined as the training set, and the rest (68 observations) are determined as the testing set.

```
set.seed(10)
train = rep(FALSE, nrow(dat1))
train[sample(1:nrow(dat1), round(nrow(dat1)*0.9))] = TRUE

m = matrix(c(1:10,rep(11,5)),nrow = 3, ncol = 5, byrow = TRUE)
layout(mat = m,heights = c(0.4,0.4,0.2))
dat1$case = as.numeric(dat1$case)-1
for(j in 1:10){
  par(mar = c(4,2,1,1))
  plot(density(dat1[train,j]), col = "#D6604D", lwd = 2, main = "", xlab = cols[j])
  lines(density(dat1[,j]), col = "#2166AC", lwd = 2)
}
dat1$case = factor(dat1$case, levels = 0:1)
plot(1, type = "n", axes=FALSE, xlab="", ylab="")
plot_colors = c("#D6604D","#2166AC")
legend(x = "top",inset = 0,
        legend = c("train", "full"),
        col=plot_colors, lwd=2, horiz = TRUE, bty =  "n")
```

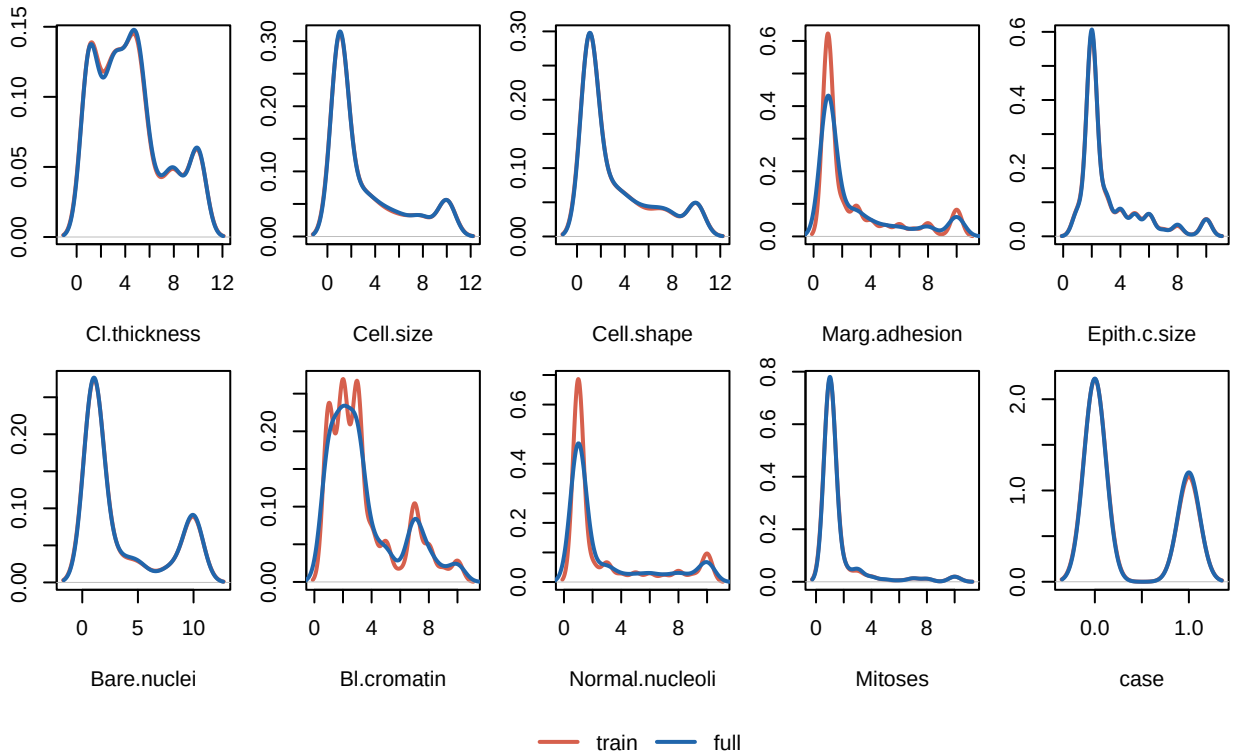


Figure 2: The resulting distribution of each predictor after splitting

Figure 2 shows the distribution of each variable in the training set and the testing set. The red curves refer to the density curves in the training set and the blue ones refer to those in the full (original) set. Overall, the red curves and the blue curves look very similar to each other for each variable, implying that the splittng of the data is uniform enough so that no obvious destruction on the data structure is present.

B. Notations

During the model fitting process, $x_1, x_2, \dots, x_9$ are used to denote the predictors (the variables except for case). As for the case, the response variable, it is denoted by Y. In addition, $\pi = P(Y = 1 \,|\, x_1, x_2, \dots, x_9)$ denotes the conditional probability of malignant given certain predictors.

```
fit.res = matrix(NA, nrow = 6, ncol = 4)
colnames(fit.res) = c("Model", "Accuracy", "FP", "FN")
fit.res[,1] = c("Logistic regression", "LDA", "QDA", "Gaussian NB", "Nonparametric NB", "KNN")
```

2.2.1 Logistic regression

First, consider the model including only the main effect of each predictor, that is

$$log\Big(\frac{\pi}{1-\pi}\Big) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9 \tag{1}$$

```
logit.fit = glm(case~. , data = dat1, subset = train ,family=binomial)
summ = as.data.frame(round(summary(logit.fit)$coefficients,4))
summkable = function(summ, cap){
  kable(summ,
  format = "latex", booktabs = TRUE, col.names = c("Estimate","SE","t-value","p-value"), align = "c",
  escape = FALSE, linesep = "", caption = cap) %>%
  kable_styling(latex_options = c("striped", "hold_position"))
}
summkable(summ, "Summary for Model $\\ref{eq:logit}$")
```

Table 8: Summary for Model 1

	Estimate	SE	t-value	p-value
(Intercept)	-9.8189	1.1697	-8.3946	0.0000
Cl.thickness	0.5342	0.1464	3.6483	0.0003
Cell.size	0.0271	0.2124	0.1277	0.8984
Cell.shape	0.2510	0.2424	1.0356	0.3004
Marg.adhesion	0.3304	0.1241	2.6625	0.0078
Epith.c.size	0.0837	0.1610	0.5200	0.6031
Bare.nuclei	0.3881	0.0956	4.0588	0.0000
Bl.cromatin	0.4575	0.1849	2.4740	0.0134
Normal.nucleoli	0.2134	0.1138	1.8762	0.0606
Mitoses	0.3617	0.3104	1.1653	0.2439

According to Table 8, some of the variables are not significant under significance level $\alpha = 0.05$. Hence some of the variables should be removed from the model. The variable selection procedure adopted here is to remove the variable with the least significance in terms of its p-value step by step. Notice that in each step only "one" variable would be removed. Such procedure is conducted until all the remaining variables are significant, that is all the p-values are smaller than 0.05.

```
logit.fit = update(logit.fit, .~.-Cell.size)
logit.fit = update(logit.fit, .~.-Epith.c.size)
logit.fit = update(logit.fit, .~.-Mitoses)
logit.fit = update(logit.fit, .~.-Cell.shape)
summ = as.data.frame(round(summary(logit.fit)$coefficients,4))
summkable(summ, "Summary for revised Model $\\ref{eq:logit}$")
```

Table 9: Summary for revised Model 1

	Estimate	SE	t-value	p-value
(Intercept)	-10.0218	1.1409	-8.7839	0.0000
Cl.thickness	0.6882	0.1328	5.1825	0.0000
Marg.adhesion	0.3877	0.1193	3.2485	0.0012
Bare.nuclei	0.4405	0.0890	4.9498	0.0000
Bl.cromatin	0.5921	0.1673	3.5389	0.0004
Normal.nucleoli	0.3299	0.0991	3.3281	0.0009

Table 9 shows the summary for the revised fitted model. Cl.thickness, Marg.adhesion, Bare.nuclei, Bl.cromatin and Normal.nucleoli are those still staying in the model when the procedure ends. Recall that

in EDA we have talked about the potential benefit of adding interaction terms into the model. Hence, we further revise and refit the model by adding all possible interaction terms of Cl.thickness, Marg.adhesion, Bare.nuclei, Bl.cromatin and Normal.nucleol into it (see Model (2)). Again, the procedure just mentioned is also applied here in order to remove the nonsignificant interaction terms.

$$log\Big(\frac{\pi}{1-\pi}\Big) = \alpha_0 + \alpha_1\, x_1 + \alpha_4\, x_4 + \alpha_6\, x_6 + \alpha_7\, x_7 + \alpha_8\, x_8 + \alpha_{14}\, x_1 x_4 + \alpha_{46}\, x_4 x_6 \tag{2}$$

```
logit.fit = glm(case~(Cl.thickness+Marg.adhesion+Bare.nuclei+Bl.cromatin+Normal.nucleoli)^2,
                data = dat1, subset = train ,family=binomial)
logit.fit = update(logit.fit, .~.-Cl.thickness:Bl.cromatin)
logit.fit = update(logit.fit, .~.-Bare.nuclei:Bl.cromatin)
logit.fit = update(logit.fit, .~.-Marg.adhesion:Bl.cromatin)
logit.fit = update(logit.fit, .~.-Cl.thickness:Normal.nucleoli)
logit.fit = update(logit.fit, .~.-Marg.adhesion:Normal.nucleoli)
logit.fit = update(logit.fit, .~.-Cl.thickness:Bare.nuclei)
logit.fit = update(logit.fit, .~.-Bl.cromatin:Normal.nucleoli)
logit.fit = update(logit.fit, .~.-Bare.nuclei:Normal.nucleoli)
summ = as.data.frame(round(summary(logit.fit)$coefficients,4))
rownames(summ)[7:8] = c("$\\text{Cl.thickness} \\times \\text{Marg.adhesion}$",
                        "$\\text{Marg.adhesion} \\times \\text{Bare.nuclei}$")
summkable(summ, "Summary for revised Model $\\ref{eq:logit2}$")
```

Table 10: Summary for revised Model 2

	Estimate	SE	t-value	p-value
(Intercept)	-13.0542	1.7953	-7.2715	0.0000
Cl.thickness	1.0478	0.2267	4.6212	0.0000
Marg.adhesion	1.2618	0.3163	3.9892	0.0001
Bare.nuclei	0.7164	0.1602	4.4719	0.0000
Bl.cromatin	0.6112	0.1835	3.3315	0.0009
Normal.nucleoli	0.3189	0.1029	3.0981	0.0019
Cl.thickness × Marg.adhesion	-0.1090	0.0451	-2.4160	0.0157
Marg.adhesion × Bare.nuclei	-0.0740	0.0316	-2.3446	0.0190

Table 10 shows that besides Cl.thickness, Marg.adhesion, Bare.nuclei, Bl.cromatin and Normal.nucleoli, the interaction terms Cl.thickness × Marg.adhesion and Marg.adhesion × Bare.nuclei also seem to be useful in predicting the case in the sense that the p-values of these two interaction terms are smaller than 0.05.

```
pred = predict(logit.fit, newdata = dat1[!train,], type = "response")
rocobj = roc(dat1[!train,]$case, pred)
threshold = rocobj$thresholds[which.max(rocobj$sensitivities + rocobj$specificities)]
auc.value = round(auc(dat1[!train,]$case, pred), 4)
ggroc(rocobj, colour = 'steelblue', size = 2) +
  ggtitle(paste0('ROC Curve ', '(AUC = ', auc.value, ')')) +
  theme_minimal()
```

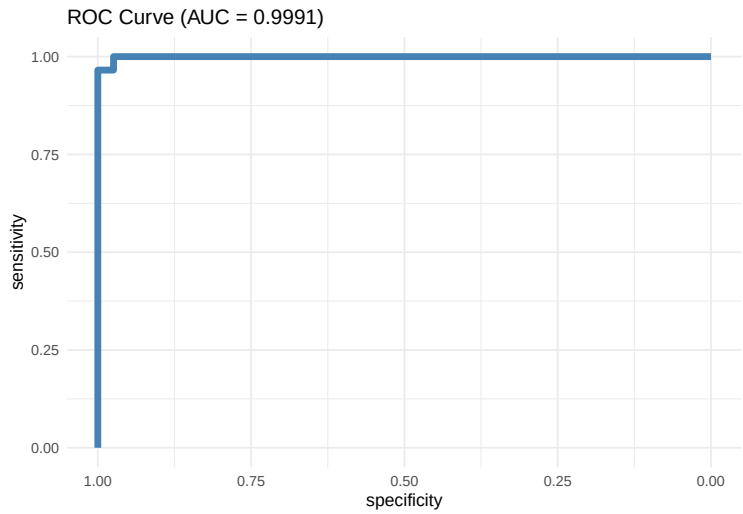


Figure 3: ROC for fitted logistic regression

```
pred.class = rep(0, length(pred)); pred.class[pred >= threshold] = 1
confusion = table(pred.class, dat1[!train,]$case)
fit.res[1, 2:4] = round(c(sum(diag(confusion))/sum(confusion),
                 confusion[2, 1]/sum(confusion[,1]),
                 confusion[1, 2]/sum(confusion[,2])))
```

Figure 3 shows the ROC for the fitted logistic regression. The AUC is almost 1, and the optimal threshold 0.0717 is obtained by finding that maximizes the sum of sensitivity and specificity.

2.2.2 Discriminant analysis

In this part, four kinds of discriminant analysis are conducted, including linear discriminant analysis, quadratic discriminant analysis, Gaussian naive Bayes classifier and nonparametric naive Bayes classifier.

1. Linear discriminant analysis

Inspired by the fitting result of the logistic regression, Cl.thickness, Marg.adhesion, Bare.nuclei, Bl.cromatin, Normal.nucleoli, Cl.thickness × Marg.adhesion and Marg.adhesion × Bare.nuclei are used as the predictor in the following discriminant analysis.

```
lda.fit=lda(case~Cl.thickness+Marg.adhesion+Bare.nuclei+Bl.cromatin+Normal.nucleoli+
              Cl.thickness:Marg.adhesion+Marg.adhesion:Bare.nuclei, data=dat1, subset = train)
```

```
prior = t(data.frame(lda.fit$prior))
rownames(prior) = NULL
kable(
  round(prior,2), format = "latex", longtable = TRUE, booktabs = TRUE,
  col.names = c("Benign", "Malignant"),
  align = "c", caption = "Prior probabilities for Benign and Malignant", escape = FALSE
  )
```

Table 11: Prior probabilities for Benign and Malignant

Benign	Malignant
0.66	0.34

Table 11 shows the prior probabilities for the benign and the malignant class. The prior probability for the benign is almost as twice as that for the malignant, letting the classifier built on linear discriminant analysis favor the benign class more.

```
grp.means = data.frame(lda.fit$means)
colnames(grp.means)[6:7] = c("$\\text{Cl.thickness} \\times \\text{Marg.adhesion}$",
                        "$\\text{Marg.adhesion} \\times \\text{Bare.nuclei}$")
rownames(grp.means) = c("Benign", "Malignant")
kable(
  round(grp.means[,1:3],2), format = "latex", longtable = TRUE, booktabs = TRUE,
  align = "c", caption = "Group means for each variable", escape = FALSE
  )  %>%
  kable_styling(latex_options = c("striped"))
```

Table 12: Group means for each variable

	Cl.thickness	Marg.adhesion	Bare.nuclei
Benign	2.95	1.35	1.35
Malignant	7.21	5.56	7.69

```
kable(
  round(grp.means[,4:7],2), format = "latex", longtable = TRUE, booktabs = TRUE,
  align = "c", caption = "Group means for each variable", escape = FALSE
  ) %>%
  kable_styling(latex_options = c("striped"))
```

Table 13: Group means for each variable

	Bl.cromatin	Normal.nucleoli	Cl.thickness × Marg.adhesion	Marg.adhesion × Bare.nuclei
Benign	2.09	1.27	4.38	2.26
Malignant	5.96	5.90	39.07	44.80

Table 12 and 13 shows the means for each predictor under the benign and the malignant class. The means under the malignant are all greater than those under the benign, implying that the higher the values of the predictors are, the higher the probability of the malignant class is.

```
pred = predict(lda.fit, newdata = dat1[!train,])
rocobj = roc(dat1[!train,]$case, pred$posterior[,2])
threshold = rocobj$thresholds[which.max(rocobj$sensitivities + rocobj$specificities)]
auc.value = round(auc(dat1[!train,]$case, pred$posterior[,2]), 2)
ggroc(rocobj, colour = 'steelblue', size = 2) +
  ggtitle(paste0('ROC Curve ', '(AUC = ', auc.value, ')')) +
  theme_minimal()
```

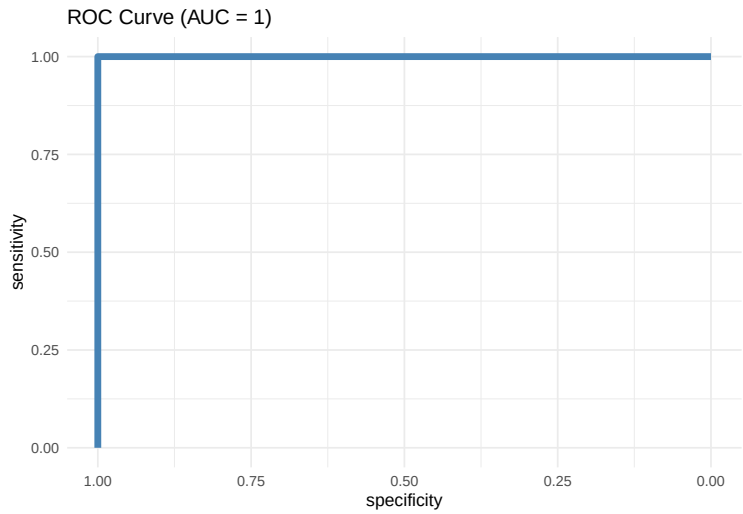


Figure 4: ROC for fitted LDA

```
pred.class = rep(0, length(pred$posterior[,2]));
pred.class[pred$posterior[,2] >= threshold] = 1
confusion = table(pred.class, dat1[!train,]$case)
fit.res[2, 2:4] = round(c(sum(diag(confusion))/sum(confusion),
                 confusion[2, 1]/sum(confusion[,1]),
                 confusion[1, 2]/sum(confusion[,2])))
```

Figure 4 shows the ROC for the fitted LDA. The AUC is almost 1, and the optimal threshold 0.0026 is obtained by finding that maximizes the sum of sensitivity and specificity.

2. Quadratic discriminant analysis

Similarly, the model of quadratic discriminant analysis which relaxes the assumption of constant covariance matrix is also fitted to the data with the same set of predictors used in the linear discriminant analysis.

```
qda.fit=lda(case~Cl.thickness+Marg.adhesion+Bare.nuclei+Bl.cromatin+Normal.nucleoli+
              Cl.thickness:Marg.adhesion+Marg.adhesion:Bare.nuclei, data=dat1, subset = train)
rocobj = roc(dat1[!train,]$case, pred$posterior[,2])
threshold = rocobj$thresholds[which.max(rocobj$sensitivities + rocobj$specificities)]
auc.value = round(auc(dat1[!train,]$case, pred$posterior[,2]), 2)
ggroc(rocobj, colour = 'steelblue', size = 2) +
  ggtitle(paste0('ROC Curve ', '(AUC = ', auc.value, ')')) +
  theme_minimal()
```

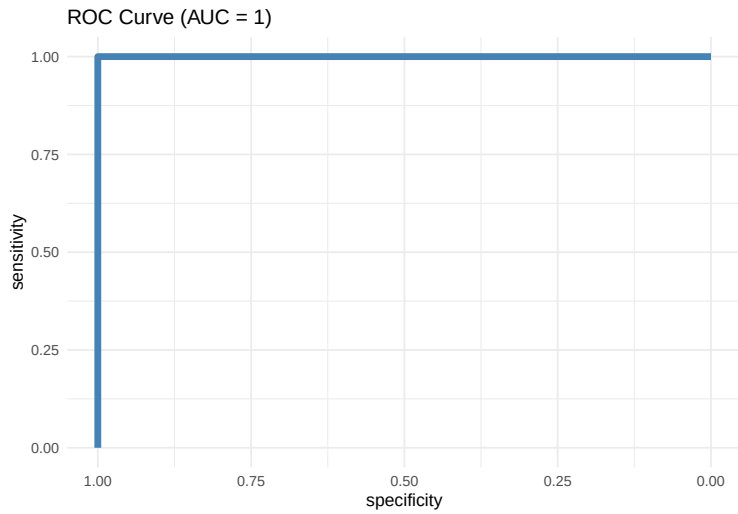


Figure 5: ROC for fitted QDA

```
pred.class = rep(0, length(pred$posterior[,2]));
pred.class[pred$posterior[,2] >= threshold] = 1
confusion = table(pred.class, dat1[!train,]$case)
fit.res[3, 2:4] = round(c(sum(diag(confusion))/sum(confusion),
                 confusion[2, 1]/sum(confusion[,1]),
                 confusion[1, 2]/sum(confusion[,2])))
```

Figure 5 shows the ROC for the fitted QDA. The AUC is almost 1, and the optimal threshold 0.0026 is obtained by finding that maximizes the sum of sensitivity and specificity.

3. Gaussian naive Bayes classifier

Considering that the number of predictors may be a little bit too many, Gaussian naive Bayes classifier is also given a try. Since interaction terms are not allowed in Bayes classifier, only Cl.thickness, Marg.adhesion, Bare.nuclei, Bl.cromatin and Normal.nucleoli are used as the predictors.

```
nb.fit = naiveBayes(case~Cl.thickness+Marg.adhesion+Bare.nuclei+Bl.cromatin+Normal.nucleoli, data=dat1,
pred.class = predict(nb.fit, newdata = dat1[!train,])
confusion = table(pred.class, dat1[!train,]$case)
fit.res[4, 2:4] = round(c(sum(diag(confusion))/sum(confusion),
                 confusion[2, 1]/sum(confusion[,1]),
                 confusion[1, 2]/sum(confusion[,2])))
```

4. Nonparametric naive Bayes classifier

As discussed in EDA, since normality assumption may not be very suitable for this data, nonparametric naive Bayes classifier is also given a try. For the same reason as Gaussian naive Bayes classifier, only Cl.thickness, Marg.adhesion, Bare.nuclei, Bl.cromatin and Normal.nucleoli are used as the predictors.

```
non.nb.fit = nonparametric_naive_bayes(dat1[train, c(1, 4, 6 ,7 , 8)], dat1[train,]$case)
pred.class = predict(non.nb.fit, newdata = as.matrix(dat1[!train, c(1, 4, 6 ,7 , 8)]))
confusion = table(pred.class, dat1[!train,]$case)
fit.res[5, 2:4] = round(c(sum(diag(confusion))/sum(confusion),
                 confusion[2, 1]/sum(confusion[,1]),
                 confusion[1, 2]/sum(confusion[,2])))
```

2.2.3 k nearest neighbor

Inspired by the fitting result of the logistic regression, Cl.thickness, Marg.adhesion, Bare.nuclei, Bl.cromatin, Normal.nucleoli, Cl.thickness × Marg.adhesion and Marg.adhesion × Bare.nuclei are used as the predictor to build a k nearest neighbor classifier. To select an optimal k, 5-fold cross validation is performed. Each fold consists of 123 observations.

```
dat1$CM = dat1$Cl.thickness*dat1$Marg.adhesion
dat1$MB = dat1$Marg.adhesion*dat1$Bare.nuclei

folds = 5
size = round(sum(train)/folds)
mcr = matrix(NA, nrow = 80, ncol = folds)

set.seed(10)
# use ross validation to select optimal k
for(k in 1:80){
  for(fold in 1:folds){
    test.index = (size*(fold-1)+1):(size*fold)
    train.index = (1:sum(train))[-test.index]
    knn.pred = knn(dat1[train, c(1, 4, 6 ,7 , 8, 11:12)][train.index,],
                   dat1[train, c(1, 4, 6 ,7 , 8, 11:12)][test.index,],
                   dat1[train,][train.index,]$case, k = k)
    mcr[k, fold] = 1-sum(diag(table(knn.pred, dat1[train,][test.index,]$case)))/size
  }
}
mcr.est = c(apply(mcr,1,mean))
# mcr estimate and its CI
cv.res = matrix(c(mcr.est, mcr.est-2*mcr.est*(1-mcr.est)/sqrt(folds),
                  mcr.est+2*mcr.est*(1-mcr.est)/sqrt(folds)), ncol = 3)
matplot(cv.res, pch = 19, xlab = "k", ylab = "misclassification rate", type = "l",
        lty = c(1, 2, 2), col = "black")
k = which.min(mcr.est)
abline(v = k, col = "red")
legend("topright", c("MCR", "95% CI"), lty = 1:2)
```

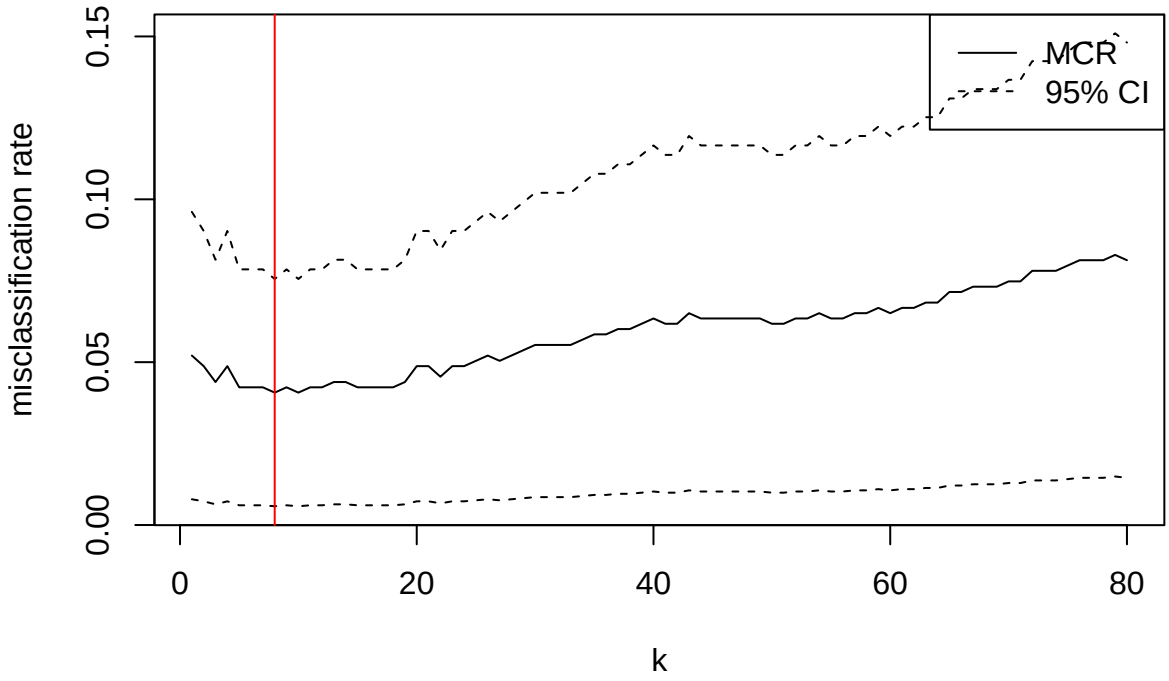


Figure 6: 5-fold cross validation for selecting k

Figure 6 shows the result of the cross validation. The black real line refers to the MCR estimate, which is calculated by averaging the MCR across the folds. The black dashed line refers to the approximated 95% confidence interval for the MCR estimate, which is calculated according to binomial distribution. The red vertical line indicate the position where the minimal MCR estimate is obtained. It turns out that k = 8 minimized the MCR estimate. Hence, we select 8 as the optimal value for k, and set it to fit the final KNN classifier.

```
knn.pred = knn(dat1[train, c(1, 4, 6 ,7 , 8, 11:12)], dat1[!train, c(1, 4, 6 ,7 , 8, 11:12)],
                 dat1[train,]$case, k = k, prob = TRUE)
confusion = table(knn.pred, dat1[!train,]$case)
fit.res[6, 2:4] = round(c(sum(diag(confusion))/sum(confusion),
                 confusion[2, 1]/sum(confusion[,1]),
                 confusion[1, 2]/sum(confusion[,2])))
```

2.3 Comparison among the different methods.

```
fit.res = data.frame(fit.res)
kable(fit.res, format = "latex", longtable = TRUE, booktabs = TRUE,
    align = "c", caption = "Summary for the prediction performance", escape = FALSE,
    linesep = "") %>%
    kable_styling(latex_options = c("striped"))
```

Table 14: Summary for the prediction performance

Model	Accuracy	FP	FN
Logistic regression	1	0	0
LDA	1	0	0
QDA	1	0	0
Gaussian NB	1	0	0
Nonparametric NB	1	0	0
KNN	1	0	0

Table 14 shows the summary for the prediction performance of each fitted model. Notice that the values in the table are taken to the fourth place after the decimal point. The probability thresholds used to classify the cases are already mentioned in the previous section for logistic regression, LDA and QDA. As for the thresholds used in Gaussian naive Bayes classifier and nonparametric naive Bayes, they are 0.5, which are the default values in `naiveBayes` and `nonparametric_naive_bayes`. One can see that all the fitted model have execellent prediction performance in the sense that their "testing" accuracies are almost 100%.

2.4 Make your conclusions on data contents.

It is found that Cl.thickness, Marg.adhesion, Bare.nuclei, Bl.cromatin and Normal.nucleoli are very useful in predicting the class. More concretely, the higher the value of each predictor predicts the higher the probability of malignant class. Furthermore, Cl.thickness $\times$ Marg.adhesion and Marg.adhesion $\times$ Bare.nuclei are found significant with negative effect in the logistic regression analysis. Compared with Figure 1, such finding is quite reasonable, since one can see that the effect of Marg.adhesion on the probability of malignant indeed decreases as Cl.thickness and Bare.nuclei increases.

3 Problem 2: Glass Data

These data consist of 214 examples of the chemical analysis of 6 different types of glass (the target class to be predicted). There are 9 chemical variables for glass classification. More data descriptions can be found by typing `help(Glass)` in r command.

This problem concerns about multi-class classification. You may use multi-class classification methods for all class simultaneously, or use 1-vs-1 or 1-vs-others schemes to build up your ensemble classifier.

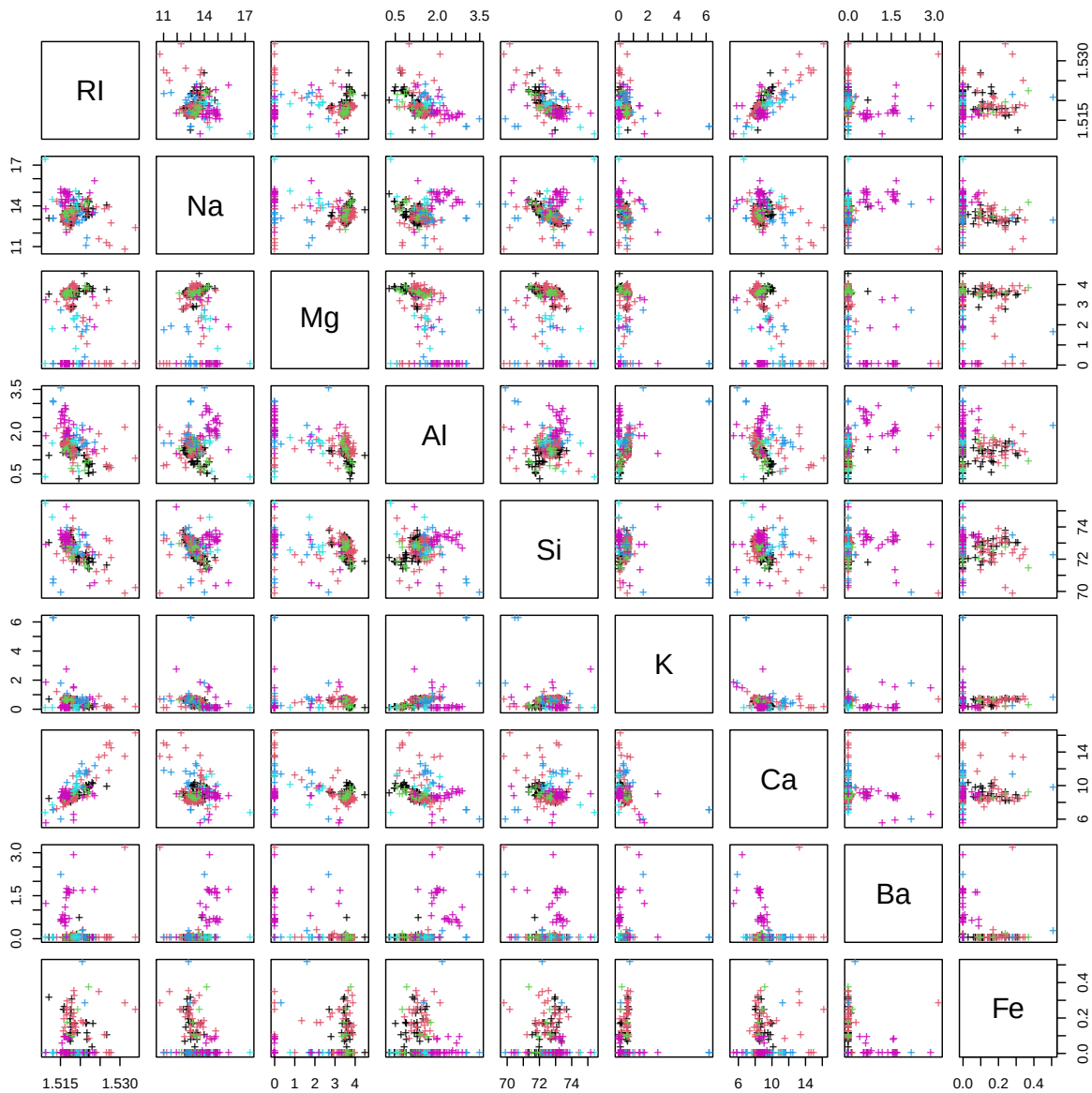
```
data(Glass)
head(Glass)
```

```
##        RI    Na   Mg   Al    Si    K   Ca Ba   Fe Type
## 1 1.52101 13.64 4.49 1.10 71.78 0.06 8.75  0 0.00    1
## 2 1.51761 13.89 3.60 1.36 72.73 0.48 7.83  0 0.00    1
## 3 1.51618 13.53 3.55 1.54 72.99 0.39 7.78  0 0.00    1
## 4 1.51766 13.21 3.69 1.29 72.61 0.57 8.22  0 0.00    1
## 5 1.51742 13.27 3.62 1.24 73.08 0.55 8.07  0 0.00    1
## 6 1.51596 12.79 3.61 1.62 72.97 0.64 8.07  0 0.26    1
```

```
#View(Glass)
summary(Glass)
```

```
##        RI              Na              Mg              Al
##  Min.   :1.511   Min.   :10.73   Min.   :0.000   Min.   :0.290
##  1st Qu.:1.517   1st Qu.:12.91   1st Qu.:2.115   1st Qu.:1.190
##  Median :1.518   Median :13.30   Median :3.480   Median :1.360
##  Mean   :1.518   Mean   :13.41   Mean   :2.685   Mean   :1.445
##  3rd Qu.:1.519   3rd Qu.:13.82   3rd Qu.:3.600   3rd Qu.:1.630
##  Max.   :1.534   Max.   :17.38   Max.   :4.490   Max.   :3.500
##        Si              K                Ca               Ba
##  Min.   :69.81   Min.   :0.0000   Min.   : 5.430   Min.   :0.000
##  1st Qu.:72.28   1st Qu.:0.1225   1st Qu.: 8.240   1st Qu.:0.000
##  Median :72.79   Median :0.5550   Median : 8.600   Median :0.000
##  Mean   :72.65   Mean   :0.4971   Mean   : 8.957   Mean   :0.175
##  3rd Qu.:73.09   3rd Qu.:0.6100   3rd Qu.: 9.172   3rd Qu.:0.000
##  Max.   :75.41   Max.   :6.2100   Max.   :16.190   Max.   :3.150
##        Fe          Type
##  Min.   :0.00000   1:70
##  1st Qu.:0.00000   2:76
##  Median :0.00000   3:17
##  Mean   :0.05701   5:13
##  3rd Qu.:0.10000   6: 9
##  Max.   :0.51000   7:29
```

```r
pairs(Glass[,1:9], col=Glass[,10], pch="+") #view data (colored by glass type)
```




```
cols = colnames(Glass)
```

3.1 Exploratory Data Analysis

3.1.1 Data Description

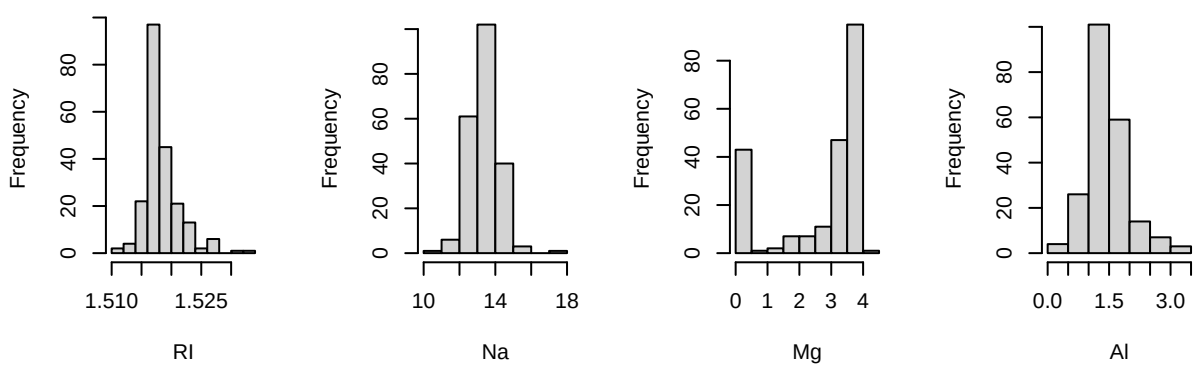
This is a dataset consisting of 214 observations on 10 variables. Nine of them is continuous variables which will be treated as the predictors. The only categorical variable is Type which will be treated as the target response. No missing value is present in the data. In order to understand the details of the distribution of each predictor, the five-number summary and the mean along with the histogram for each predictor are reported as follows.

```
summ = sapply(Glass[,1:9], summary)
kable(round(summ[,1:4], 2), format = "latex", longtable = TRUE, booktabs = TRUE,
  align = "c", caption = "Five-number-summary for each variable", escape = FALSE, linesep = ""
  )  %>%
  kable_styling(latex_options = c("striped"))
```

Table 15: Five-number-summary for each variable

	RI	Na	Mg	Al
Min.	1.51	10.73	0.00	0.29
1st Qu.	1.52	12.91	2.11	1.19
Median	1.52	13.30	3.48	1.36
Mean	1.52	13.41	2.68	1.44
3rd Qu.	1.52	13.83	3.60	1.63
Max.	1.53	17.38	4.49	3.50

```
par(mfrow = c(1,4))
for(i in 1:4){
  hist(Glass[,i], main = "", xlab = cols[i])
}
```

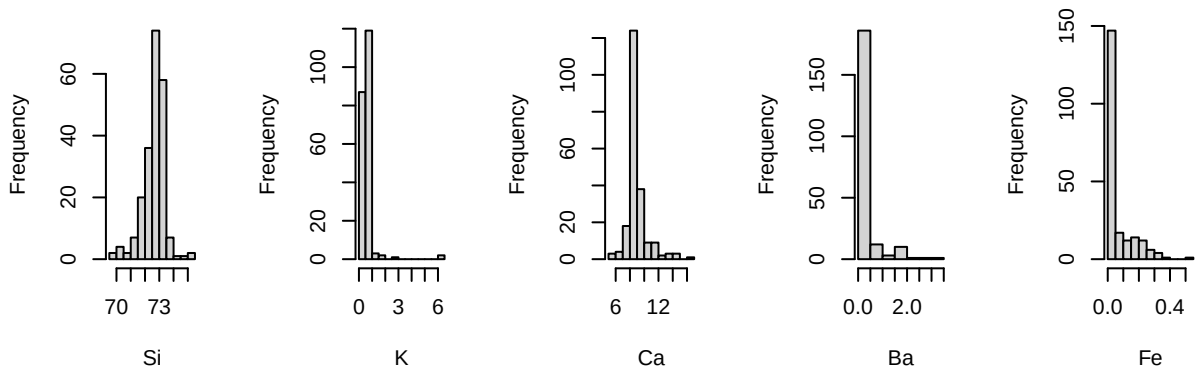



```
summ = sapply(Glass[,1:9], summary)
kable(round(summ[,5:9], 2), format = "latex", longtable = TRUE, booktabs = TRUE,
  align = "c", caption = "Five-number-summary for each variable", escape = FALSE, linesep = ""
  ) %>%
  kable_styling(latex_options = c("striped"))
```

Table 16: Five-number-summary for each variable

	Si	K	Ca	Ba	Fe
Min.	69.81	0.00	5.43	0.00	0.00
1st Qu.	72.28	0.12	8.24	0.00	0.00
Median	72.79	0.56	8.60	0.00	0.00
Mean	72.65	0.50	8.96	0.18	0.06
3rd Qu.	73.09	0.61	9.17	0.00	0.10
Max.	75.41	6.21	16.19	3.15	0.51

```
par(mfrow = c(1,5))
for(i in 5:9){
  hist(Glass[,i], main = "", xlab = cols[i])
}
```



The difference between the median and the mean shown in Table 15 and 16 suggest a left-skewed distribution for Mg, right-skewed distributions for K and Ba, and symmetric distributions for the rest. Moreover, as shown in the histogram for Mg, it seems that Mg has bimodal distributions, which may be resulted from the distinct distributions under the different types of glass.

```
counts = matrix(table(Glass$Type), nrow = 1)
kable(counts, format = "latex", longtable = TRUE, booktabs = TRUE,
      col.names = c(1, 2, 3, 5, 6, 7),align = "c", caption = "Summary for Type", escape = FALSE,
      linesep = "")
```

Table 17: Summary for Type

1	2	3	5	6	7
70	76	17	13	9	29

Table 17 shows the number of observations within each type of glass. One can see that the data does not distribute uniformly across the types. Especially in Type 5 and 6, the numbers of observations are very small. Such fact may cause problem when estimating the MCR for the Type 5 and Type 6 observations.

3.1.2 Data Feature

To explore the basic relationships among the predictors, data visualization and correlation analysis are first conducted as follows.

```
Glass$Type = factor(Glass$Type, levels = c(1, 2, 3, 5, 6, 7))
ggpairs(Glass, columns = 1:9, ggplot2::aes(colour=Type, alpha = 0.5),
        upper = list(continuous = wrap("cor", size = 1.8)),
        lower = list(continuous = wrap("points", size=0.2))) %>%
        + theme(strip.placement = "outside", text = element_text(size = 8))
```

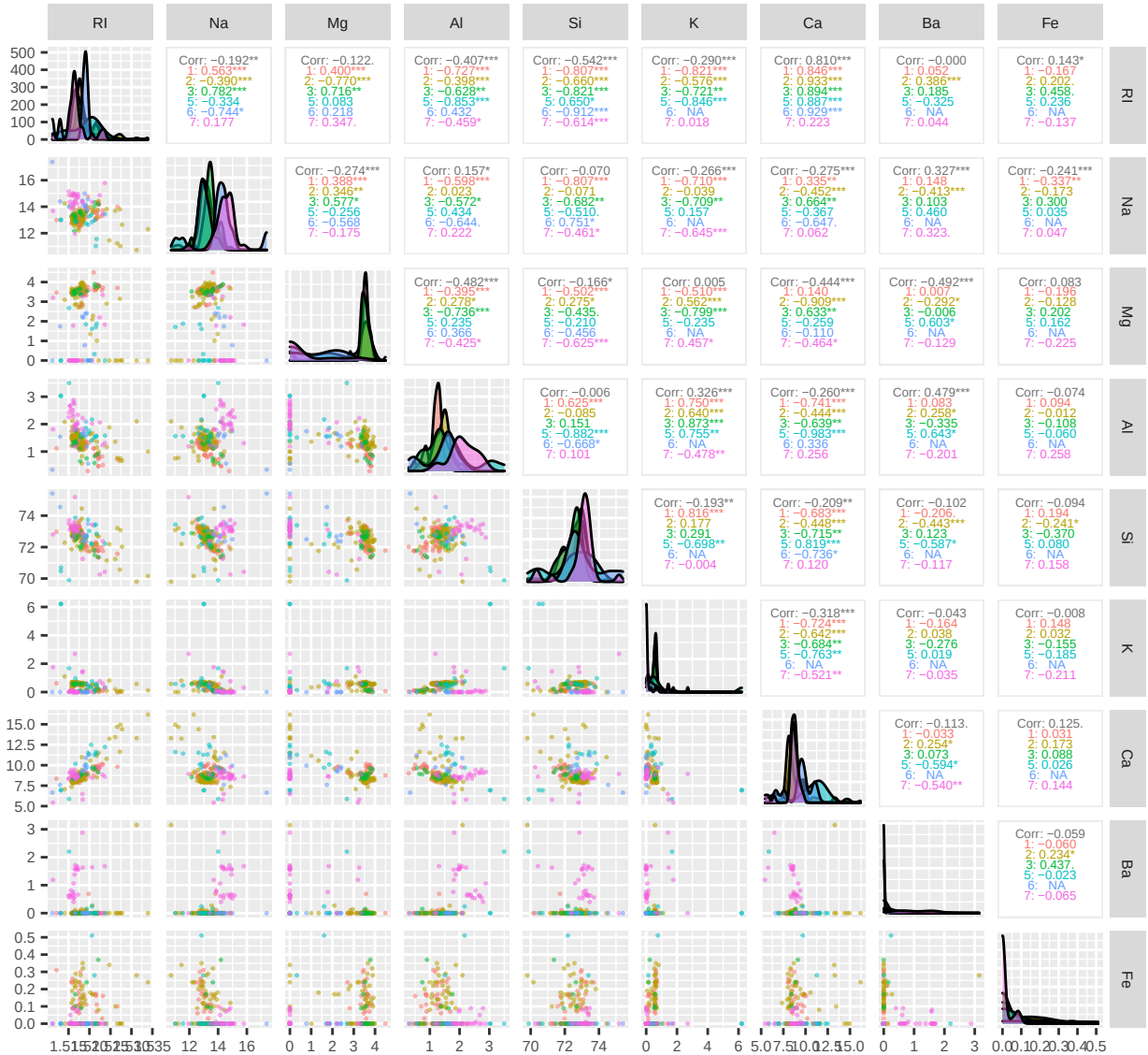


Figure 7: Summary for the relationships among the variables

In Figure 7, the types of glass (1~7) are discriminated by different colors. From the diagonal (the histograms), it seems that RI, Na, Mg, AI, Si and Ca may be useful in predicting the types in the sense that each of those has somewhat different distributions among the six types. Moreover, from the lower left (the scatter

plots) one can see that neither normality nor constant covariance matrix seem to be acceptable assumptions. Nevertheless, considering that the predictors and the categories of Type are both a lot, no strong comments would be made here.

3.2 Performing the classification task

A. Train-test splitting

To evaluate the test mean squared error, the data is splitted into a training set and a testing set. About 70% of the data (46 observations) are randomly determined as the training set, and the rest (68 observations) are determined as the testing set.

```
set.seed(5)
train = rep(FALSE, nrow(Glass))
train[sample(1:nrow(Glass), round(nrow(Glass)*0.7))] = TRUE
# Small adustment for the testing set
train[c(sample(which(Glass$Type == 3 & train), 1),
        sample(which(Glass$Type == 6 & train), 3))] = FALSE

m = matrix(c(1:10,rep(11,5)),nrow = 3, ncol = 5, byrow = TRUE)
layout(mat = m,heights = c(0.4,0.4,0.2))
Glass$Type = c(as.numeric(Glass$Type)[1:163], as.numeric(Glass$Type)[164:214]+1)
for(j in 1:10){
  par(mar = c(4,2,1,1))
  plot(density(Glass[train,j]), col = "#D6604D", lwd = 2, main = "", xlab = cols[j])
  lines(density(Glass[,j]), col = "#2166AC", lwd = 2)
}
Glass$Type = factor(Glass$Type, levels = c(1, 2, 3, 5, 6, 7))
plot(1, type = "n", axes=FALSE, xlab="", ylab="")
plot_colors = c("#D6604D","#2166AC")
legend(x = "top",inset = 0,
        legend = c("train", "full"),
        col=plot_colors, lwd=2, horiz = TRUE, bty =  "n")
```

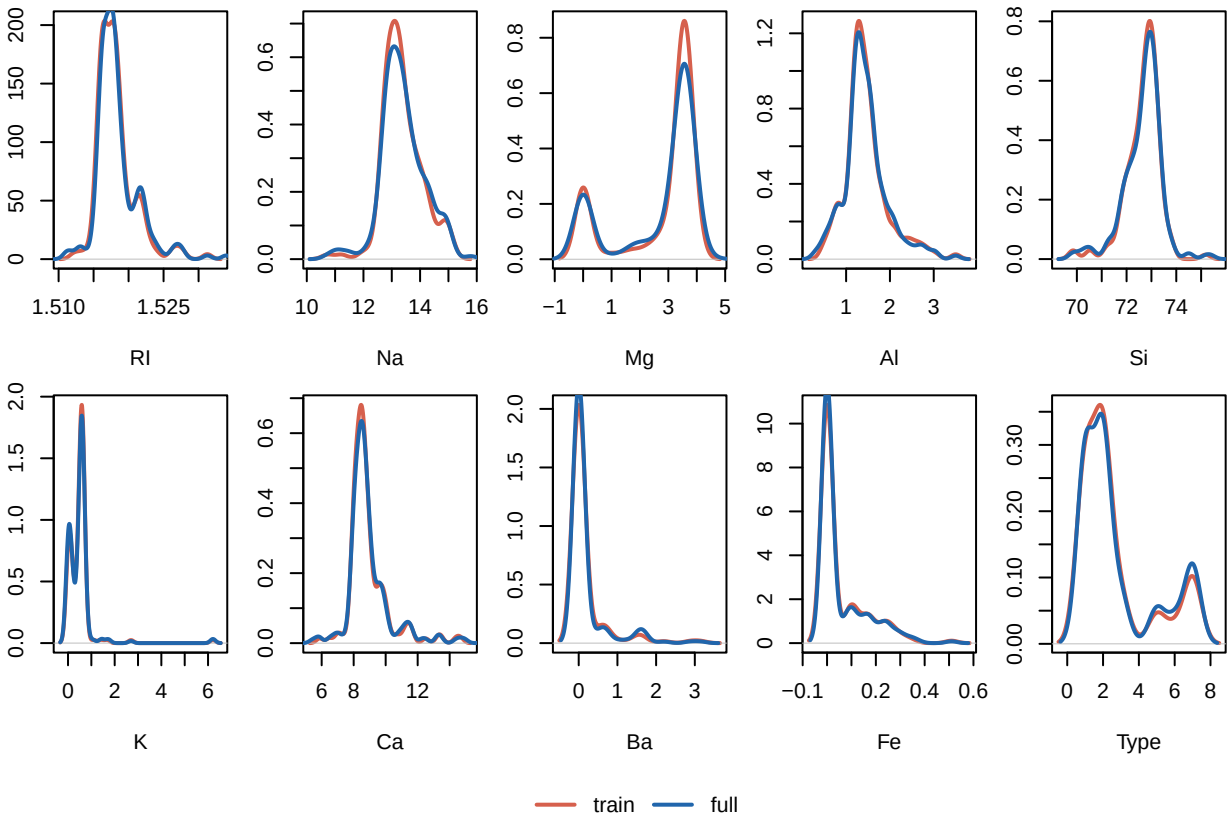


Figure 8: The resulting distribution of each predictor after splitting

Notice that the number of observations across Type in the testing set is adjusted a little bit to make sure there are at least 5 observations in each type (see Table @ref(tab:test.ype)). This is done in hope that we can obtain a smoother and more stable MCR estimate for each type.

```
counts = matrix(table(Glass[!train,]$Type), nrow = 1)
kable(counts, format = "latex", longtable = TRUE, booktabs = TRUE,
      col.names = c(1, 2, 3, 5, 6, 7), align = "c", caption = "Summary for Type in the testing set",
      escape = FALSE, linesep = "") %>%
      kable_styling(latex_options = c("hold_position"))
```

Table 18: Summary for Type in the testing set

1	2	3	5	6	7
22	20	5	5	5	11

Figure 8 shows the distribution of each variable in the training set and the testing set. The red curves refer to the density curves in the training set and the blue ones refer to those in the full (original) set. Overall, the red curves and the blue curves look very similar to each other for each variable, implying that the splittng of the data is uniform enough so that no obvious destruction on the data structure is present.

```
fit.res = matrix(NA, nrow = 5, ncol = 8)
colnames(fit.res) = c("Model", "Accuracy", paste("Type", c(1, 2, 3, 5, 6, 7), "MCR"))
fit.res[,1] = c("Multiclass LR", "LDA", "Gaussian NB", "Nonparametric NB", "KNN")
```

B. Notations

During the model fitting process, $x_1, x_2, \dots, x_9$ are used to denote the predictors. As for the target response, it is represented by $Y_1, Y_2, \dots, Y_7$, where

$$Y_i = \begin{cases} 1, & \text{if the observation is in Type i} \\ 0, & \text{otherwise} \end{cases} \qquad i = 1, 2, 3, 5, 6, 7.$$

In addition, $\pi_i = P(Y_i = 1 \,|\, x_1, x_2, \dots, x_9)$ denotes the conditional probability of Type i given certain predictors.

3.2.1 Multiclass logistic regression

According to EDA, consider the model including only the main effects of RI, Na, Mg, AI, Si and Ca that is

$$log\left(\frac{\pi_i}{\pi_1}\right) = \beta_{i0} + \beta_{i1}\, x_1 + \beta_{i2}\, x_2 + \beta_{i3}\, x_3 + \beta_{i4}\, x_4 + \beta_{i5}\, x_5 + \beta_{i7}\, x_7, \quad i = 2, 3, 5, 6, 7. \tag{3}$$

Notice that Type 1 is set to be the baseline type.

```
mlr.fit = multinom(Type~RI+Na+Mg+Al+Si+Ca, data = Glass, subset = train)
coefs = coef(mlr.fit)
ses = summary(mlr.fit)$standard.errors
summs = list()
for(i in c(2, 3, 5, 6, 7)){
  summ = data.frame(est = coefs[rownames(coefs) == i,],
                    se = ses[rownames(coefs) == i,],
                    z = coefs[rownames(coefs) == i,]/ses[rownames(coefs) == i,])
  summ$p = (1 - pnorm(abs(summ$z), 0, 1)) * 2
  summs[[i]] = summ
}
```

```
summ = summs[[2]]
for(s in 3:length(summs)){
  summ = rbind(summ, summs[[s]])
}
kable(round(summ, 4),
  format = "latex", booktabs = TRUE, col.names = c("Estimate","SE","z-value","p-value"), align = "c",
  escape = FALSE, linesep = "", caption = "Summary for Model \\ref{eq:mlr1}") %>%
  kable_styling(latex_options = c("striped", "hold_position")) %>%
  group_rows("i = 2", 1, 7) %>%
  group_rows("i = 3", 8, 14) %>%
  group_rows("i = 5", 15, 21) %>%
  group_rows("i = 6", 22, 28) %>%
  group_rows("i = 7", 29, 35)
```

Table 19 shows the summary for fitted model (3). It is found that RI, Na, Mg, AI, Si and Ca are useful in predicting types of glass. Though some of the parameters in the model are not significant under significance level $\alpha = 0.05$ according to the Wald's test, we do not need to care so much about it since our goal is mainly on prediction. Interaction terms will not be further considered to avoid problem of sparsity due to the limiting sample size.

Table 19: Summary for Model 3

	Estimate	SE	z-value	p-value
i = 2				
(Intercept)	120.6582	0.3486	346.1502	0.0000
RI	230.3281	0.6164	373.6482	0.0000
Na	-3.2761	0.5511	-5.9444	0.0000
Mg	-6.2294	0.8714	-7.1491	0.0000
Al	-1.4466	1.3942	-1.0376	0.2994
Si	-5.0087	0.1264	-39.6262	0.0000
Ca	-4.5811	0.5166	-8.8672	0.0000
i = 3				
(Intercept)1	8.8831	0.0352	252.4977	0.0000
RI1	-20.9700	0.0576	-363.9708	0.0000
Na1	1.5611	0.7153	2.1825	0.0291
Mg1	-0.0014	1.2232	-0.0012	0.9991
Al1	1.3570	1.7686	0.7673	0.4429
Si1	-0.0342	0.1690	-0.2022	0.8398
Ca1	0.1915	0.6660	0.2876	0.7737
i = 5				
(Intercept)2	56.4437	0.8297	68.0253	0.0000
RI2	68.5576	1.2601	54.4048	0.0000
Na2	-4.4511	3.4565	-1.2878	0.1978
Mg2	-12.0575	6.4615	-1.8661	0.0620
Al2	9.4495	3.3898	2.7877	0.0053
Si2	-0.5137	1.3998	-0.3670	0.7136
Ca2	-5.6208	4.3048	-1.3057	0.1916
i = 6				
(Intercept)3	-53.1637	0.0533	-998.1515	0.0000
RI3	-81.5901	0.0843	-968.3135	0.0000
Na3	4.0495	1.3501	2.9993	0.0027
Mg3	-0.7690	1.6672	-0.4613	0.6446
Al3	7.1208	3.3248	2.1417	0.0322
Si3	1.3581	0.3837	3.5400	0.0004
Ca3	1.5650	1.2080	1.2956	0.1951
i = 7				
(Intercept)4	-120.1106	0.3601	-333.5303	0.0000
RI4	-186.8678	0.5497	-339.9305	0.0000
Na4	-1.5079	4.8873	-0.3085	0.7577
Mg4	-17.2969	8.5198	-2.0302	0.0423
Al4	6.8657	11.9390	0.5751	0.5652
Si4	7.5954	1.6921	4.4887	0.0000
Ca4	-11.7112	6.3748	-1.8371	0.0662

```r
pred.prob = predict(mlr.fit, newdata = Glass[!train, 1:9], type = "prob")
pred.class = c(1, 2, 3, 5, 6, 7)[apply(pred.prob, 1, which.max)]
confusion = table(pred.class, Glass[!train,]$Type)
confusion = rbind(confusion[1:2,], rep(0, 6), confusion[3:5,])
acc = sum(diag(confusion))/sum(confusion)
mcr = round(1-diag(confusion)/apply(confusion, 2, sum), 2)
```

```
fit.res[1, 2:8] = c(acc, mcr)
```

3.2.2 Discriminant analysis

In this part, four kinds of discriminant analysis are conducted, including linear discriminant analysis, quadratic discriminant analysis, Gaussian naive Bayes classifier and nonparametric naive Bayes classifier.

1. Linear discriminant analysis

Inspired by the fitting result of the multiclass logistic regression, RI, Na, Mg, AI, Si and Ca are used as the predictor in the following discriminant analysis.

```
lda.fit=lda(Type~RI+Na+Mg+Al+Si+Ca, data=Glass, subset = train)
```

```
prior = t(data.frame(lda.fit$prior))
rownames(prior) = NULL
kable(
  round(prior,2), format = "latex", longtable = TRUE, booktabs = TRUE,
  col.names = c(1, 2, 3, 5, 6, 7),
  align = "c", caption = "Prior probabilities for the six types", escape = FALSE
  )
```

Table 20: Prior probabilities for the six types

1	2	3	5	6	7
0.33	0.38	0.08	0.05	0.03	0.12

Table 11 shows the prior probabilities for the six types. The prior probabilities for Type 1 and Type 2 are the highest, letting the classifier built on linear discriminant analysis favor Type 1 and Type 2 more than the other types.

```
grp.means = data.frame(lda.fit$means)
kable(
  round(grp.means[,1:6],2), format = "latex", longtable = TRUE, booktabs = TRUE,
  align = "c", caption = "Group means for each variable", escape = FALSE, linesep = ""
  ) %>%
  kable_styling(latex_options = c("striped"))
```

Table 21: Group means for each variable

	RI	Na	Mg	Al	Si	Ca
1	1.52	13.17	3.52	1.18	72.67	8.80
2	1.52	13.18	3.11	1.41	72.55	8.92
3	1.52	13.43	3.53	1.19	72.42	8.78
5	1.52	13.11	0.81	2.11	72.21	9.90
6	1.52	14.22	1.94	1.54	72.58	9.65
7	1.52	14.38	0.10	2.22	73.33	8.67

Table 21 shows the means for each predictor under the seven types. One may find that the means of RI within each type of glass are almost the same and suspect whether RI is really useful in prediction. No

matter what, such fact suggests that the current fitted model may not be satisfactory enough. Other DA classifiers are hence worth a try.

```
pred = predict(lda.fit, newdata = Glass[!train,])
confusion = table(pred$class, Glass[!train,]$Type)
acc = sum(diag(confusion))/sum(confusion)
mcr = round(1-diag(confusion)/apply(confusion, 2, sum), 2)
fit.res[2, 2:8] = c(acc, mcr)
```

2. Gaussian naive Bayes classifier

Considering that the number of predictors may be too many, Gaussian naive Bayes classifier is also given a try.

```
nb.fit = naiveBayes(Type~RI+Na+Mg+Al+Si+Ca, data=Glass, subset = train)
pred.class = predict(nb.fit, newdata = Glass[!train,])
confusion = table(pred.class, Glass[!train,]$Type)
acc = sum(diag(confusion))/sum(confusion)
mcr = round(1-diag(confusion)/apply(confusion, 2, sum), 2)
fit.res[3, 2:8] = c(acc, mcr)
```

4. Nonparametric naive Bayes classifier

As discussed in EDA, since normality assumption may not be very suitable for this data, nonparametric naive Bayes classifier is also given a try.

```
non.nb.fit = nonparametric_naive_bayes(Glass[train, c(1, 3, 4 ,6 ,7 ,8)], Glass[train,]$Type)
pred.class = predict(non.nb.fit, newdata = as.matrix(Glass[!train, 1:9]))
confusion = table(pred.class, Glass[!train,]$Type)
acc = sum(diag(confusion))/sum(confusion)
mcr = round(1-diag(confusion)/apply(confusion, 2, sum), 2)
fit.res[4, 2:8] = c(acc, mcr)
```

3.2.3 k nearest neighbor

Inspired by the fitting result of the multiclass logistic regression, RI, Na, Mg, AI, Si and Ca are used as the predictors to build a k nearest neighbor classifier. To select an optimal k, 5-fold cross validation is performed. Each fold consists of 49 observations.

```
folds = 5
size = round(sum(train)/folds)
mcr = matrix(NA, nrow = 80, ncol = folds)

set.seed(10)
# use ross validation to select optimal k
for(k in 1:80){
  for(fold in 1:folds){
    test.index = (size*(fold-1)+1):(size*fold)
    train.index = (1:sum(train))[-test.index]
    knn.pred = knn(Glass[train, c(1, 2, 3, 4, 5, 7)][train.index,],
                   Glass[train, c(1, 2, 3, 4, 5, 7)][test.index,],
                   Glass[train,][train.index,]$Type, k = k)
    mcr[k, fold] = 1-sum(diag(table(knn.pred, Glass[train,][test.index,]$Type)))/size
```

```
  }
}
mcr.est = c(apply(mcr,1,mean))
# mcr estimate and its CI
cv.res = matrix(c(mcr.est, mcr.est-2*mcr.est*(1-mcr.est)/sqrt(folds),
                  mcr.est+2*mcr.est*(1-mcr.est)/sqrt(folds)), ncol = 3)
matplot(cv.res, pch = 19, xlab = "k", ylab = "misclassification rate", type = "l",
        lty = c(1, 2, 2), col = "black")
k = which.min(mcr.est)
abline(v = k, col = "red")
legend("topright", c("MCR", "95% CI"), lty = 1:2)
```

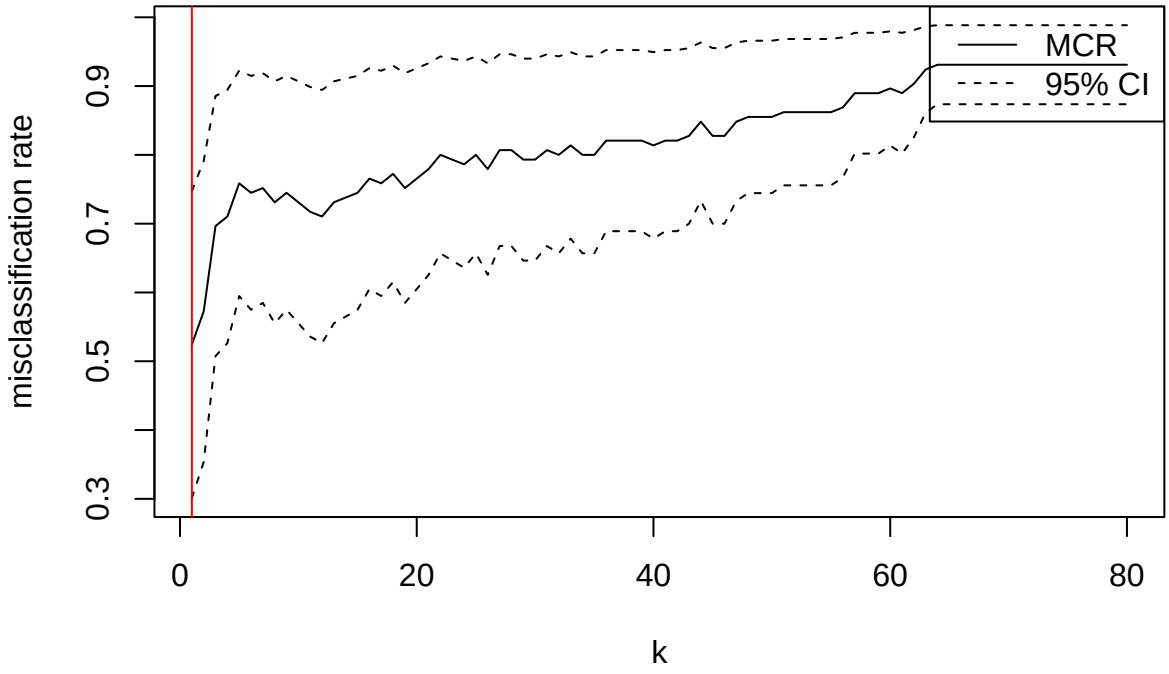


Figure 9: 5-fold cross validation for selecting k

Figure 6 shows the result of the cross validation. The black real line refers to the MCR estimate, which is calculated by averaging the MCR across the folds. The black dashed line refers to the approximated 95% confidence interval for the MCR estimate, which is calculated according to binomial distribution. The red vertical line indicate the position where the minimal MCR estimate is obtained. It turns out that k = 1 minimized the MCR estimate. Hence, we select 1 as the optimal value for k, and set it to fit the final KNN classifier.

```
knn.pred = knn(Glass[train, c(1, 2, 3, 4, 5, 7)], Glass[!train, c(1, 2, 3, 4, 5, 7)],
               Glass[train,]$Type, k = k, prob = TRUE)
confusion = table(knn.pred, Glass[!train,]$Type)
```

```
acc = sum(diag(confusion))/sum(confusion)
mcr = round(1-diag(confusion)/apply(confusion, 2, sum), 2)
fit.res[5, 2:8] = c(acc, mcr)
```

3.3 Comparison among the different methods.

```
fit.res = data.frame(fit.res)
fit.res[, 2:8] = sapply(fit.res[, 2:8], as.numeric)
fit.res[, 2:8] = round(fit.res[, 2:8], 4)
kable(fit.res[,1:4], 2, format = "latex", longtable = TRUE, booktabs = TRUE,
    align = "c", caption = "Summary for the prediction performance", escape = FALSE,
    linesep = "") %>%
    kable_styling(latex_options = c("striped"))
```

Table 22: Summary for the prediction performance

Model	Accuracy	Type.1.MCR	Type.2.MCR
Multiclass LR	0.62	0.18	0.35
LDA	0.56	0.41	0.30
Gaussian NB	0.44	0.27	0.70
Nonparametric NB	0.60	0.09	0.55
KNN	0.71	0.23	0.15

```
kable(fit.res[5:8], 2, format = "latex", longtable = TRUE, booktabs = TRUE,
    align = "c", caption = "Summary for the prediction performance (cont.)", escape = FALSE,
    linesep = "") %>%
    kable_styling(latex_options = c("striped"))
```

Table 23: Summary for the prediction performance (cont.)

Type.3.MCR	Type.5.MCR	Type.6.MCR	Type.7.MCR
1.0	0.4	0.8	0.36
1.0	0.4	0.6	0.45
1.0	0.6	0.8	0.55
1.0	0.0	0.8	0.45
0.6	0.0	0.8	0.45

Table 22 and 23 show the summary for the prediction performance of each fitted model. Notice that the values in the table are taken to the fourth place after the decimal point. One can see that KNN performs the best in terms of the "testing" accuracy and the MCRs for Type 2, 3 and 5. Moreover, nonparametric naive Bayes classifier, LDA or multiclass logistic regression may be recommended respectively for the application where Type 1, Type 6 or Type 7 sensitivities are of big concern.

3.4 Make your conclusions on data contents.

It is found that RI, Na, Mg, AI, Si and Ca are useful in predicting the types of glass. As what have been investigated, the overall prediction power is strongest when KNN is fitted with k = 1.

3.5 Remark

QDA was not conducted here since the sample size for some group is too small.